

6.8 Thermal characteristics

The operating junction temperature T_J must never exceed the maximum given in [Table 24: General operating conditions](#).

The maximum junction temperature in °C that the device can reach if respecting the operating conditions, is:

$$T_J(\text{max}) = T_A(\text{max}) + P_D(\text{max}) \times \Theta_{JA}$$

where:

- $T_A(\text{max})$ is the maximum operating ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D = P_{\text{INT}} + P_{\text{I/O}}$,
 - P_{INT} is power dissipation contribution from product of I_{DD} and V_{DD}
 - $P_{\text{I/O}}$ is power dissipation contribution from output ports where
 $P_{\text{I/O}} = \sum (V_{\text{OL}} \times I_{\text{OL}}) + \sum ((V_{\text{DDIO1}} - V_{\text{OH}}) \times I_{\text{OH}})$, taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 71. Thermal characteristics

Symbol	Parameter	Package ⁽¹⁾	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient	TSSOP20	87.2	°C/W
		UFQFPN28	75.9	
		UFQFPN32	41.8	
		LQFP32	60.5	
		UFQFPN48	31.2	
		LQFP48	60.5	
Θ_{JB}	Thermal resistance junction-board	TSSOP20	53.2	°C/W
		UFQFPN28	41.3	
		UFQFPN32	23.6	
		LQFP32	38	
		UFQFPN48	15.7	
		LQFP48	38	
Θ_{JC}	Thermal resistance junction-case	TSSOP20	24.8	°C/W
		UFQFPN28	38.4	
		UFQFPN32	22.6	
		LQFP32	21	
		UFQFPN48	13.4	
		LQFP48	21	

1. Refer to [Section 6: Package information](#) for package dimensions

6.8.1 Reference documents

[1] *Integrated Circuits Thermal Test Method Environmental Conditions - Natural Convection (Still Air)* (JESD51-2A), JEDEC, January 2008. Available from www.jedec.org.